Implication of HSAT2 Repeat Sequence RNAs in ME/CFS and Long COVID: A Hypothesis

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Human satellite 2 (HSAT2) is a pericentromeric repeat representing approximately 1.5% of the human genome. Under physiological conditions, it is transcriptionally silenced through DNA methylation and H3K9me3-dependent heterochromatin [1, 2]. In cancer, however, HSAT2 may become aberrantly expressed and promotes immune suppression through extracellular vesicle-mediated reprogramming of myeloid cells, leading notably to the expansion of myeloid-derived suppressor cells and immune exhaustion [3].

Here we propose that an analogous mechanism may contribute to the development of post-acute infection syndromes (PAIS), including Long COVID and ME/CFS. This model provides a potential explanation for the frequent observation that these disorders emerge following reactivation of latent herpesviruses. Indeed, HSAT2 RNAs are also induced upon herpesvirus replication. Herpesvirus infections are known to disrupt centromere organization and can induce transcription of pericentromeric satellite repeats, including HSAT2 [4–6]. Following repeated or closely spaced viral infections, cumulative changes in chromatin composition and loss of CpG methylation at pericentromeric loci may become sufficiently extensive to initiate a self-sustaining process of progressive epigenetic erosion [1, 7–10].

By analogy with cancer, we hypothesize that HSAT2 RNAs released in extracellular vesicles could propagate this altered epigenetic state, driving persistent immune dysregulation and immune exhaustion [3]. Rather than presenting definitive proof, this Perspective assembles the available evidence into a coherent mechanistic framework linking the initiating trigger to downstream molecular events and clinical manifestations. It also generates specific, experimentally testable predictions that can now be tested using patient-derived samples [11, 12].

1 What HSAT2 Is and Why It Is Normally Silent

HSAT2 (human satellite 2) is a stretch of DNA repeated many times in the genome, occupying approximately 1.5% of our DNA. It sits in pericentromeric regions: the DNA flanking each chromosome's central pinch point (the centromere, where the two arms meet). Repetitive sequences account for roughly half of the human genome—far more than the ∼2% that encodes proteins—yet have historically been neglected.

In healthy cells, HSAT2 does not produce RNA. This silencing is maintained by two locks: (i) DNA methylation, in which small methyl groups are added to specific sites (CpG, cytosine–phosphate–guanine dinucleotides), and (ii) tightly packed chromatin called heterochromatin, reinforced by a chemical mark on the histone proteins around which DNA is wound (H3K9me3: histone H3 trimethylated at lysine 9). Together these prevent the cellular machinery from transcribing HSAT2 into RNA [1, 2].

2 What Happens in Cancer

In several cancers, HSAT2 escapes silencing and is transcribed into RNA, which is packaged into extracellular vesicles (EVs)—tiny membrane-wrapped particles, roughly 100 nm wide, that cells shed into body fluids and travel via the bloodstream. When HSAT2-laden EVs are taken up by immature cells of the myeloid lineage (precursors of several innate immune cell types), they reprogram those cells to dampen immunity. Specifically, the EVs drive expansion of myeloid-derived suppressor cells (MDSCs)—a population of incompletely matured cells that suppress other immune cells and expand in disease.

MDSCs suppress natural killer (NK) cells—rapid killers of virus-infected and cancerous cells—and drive $CD8^+$ T-cells ("killer" T-cells critical for clearing infections) into exhaustion, a state of progressive functional loss. This is mediated by the signalling molecules IL-10, IL-35, the enzyme IDO1, and TGF-β (transforming growth factor beta) [3]. Tumours expressing HSAT2 are immunologically cold: they show little immune-cell infiltration, consistent with HSAT2 actively suppressing anti-tumour immunity [13].

Critically, cells taking up HSAT2-containing EVs produce and release their own HSAT2-carrying EVs, spreading the signal to additional cells—a self-sustaining loop that can outlast the original cancer trigger [3].

3 Extension to Post-Acute Infection Syndromes

We propose that this same chain of events may operate in post-acute infection syndromes (PAIS)—most notably Long COVID and ME/CFS. These conditions often follow a viral illness and many patients experience reactivation of dormant herpesviruses (HSV-1, cold-sore virus; HCMV, cytomegalovirus; HHV-6).

The proposed sequence:

Trigger. When a herpesvirus reactivates, it replicates at the centromeres—the pinch points of chromosomes—where viral proteins cause the local DNA to amplify abnormally [4]. This physical disruption destabilises the adjacent pericentromeric regions and switches on HSAT2 transcription [5, 6].

Cumulative erosion. Each reactivation damages chromatin structure and strips the methylation marks keeping HSAT2 silent. After repeated or closely spaced infections, cumulative loss can cross a threshold: HSAT2 production becomes self-sustaining and the epigenetic erosion continues without further viral stimulation [1, 7–10]. DNA methylation patterns are known to be altered in ME/CFS and Long COVID, and pericentromeric repeats are especially susceptible to methylation loss.

Propagation. By analogy with cancer, HSAT2 RNAs from cells experiencing centromere disruption are released into the bloodstream in EVs, reach immature myeloid cells, drive MDSC expansion, and induce T-cell exhaustion—contributing to the NK-cell dysfunction and broader immune dysregulation documented in ME/CFS and Long COVID [3]. Because EV-mediated propagation is self-sustaining, immune exhaustion could persist long after the triggering infection has cleared.

4 Testable Predictions

The model generates four predictions testable in patient samples:

(1) **EV HSAT2 elevation.** HSAT2 RNA or DNA in blood-derived EVs will be elevated in a subgroup of ME/CFS and Long COVID patients with a confirmed viral trigger.

(2) **Patient subgroups.** Patients who test positive for EV HSAT2 will show a distinct clinical profile: viral-onset history, expanded MDSC populations, and impaired NK-cell function.

(3) **Biomarker panels.** A combined panel (EV HSAT2, MDSC frequency, cytokines) will identify the affected subgroup with high specificity.

(4) **Therapeutic response.** If the hypothesis is correct, methyl donors—compounds supplying DNA methylation material (SAMe [S-adenosylmethionine], methylfolate, betaine)—will reduce EV HSAT2 more than placebo. This prediction awaits testing and is not a treatment recommendation; demethylating agents (HDAC inhibitors, 5-azacitidine) should not be administered on the basis of this model alone.

These predictions can be pursued immediately. Two HSAT2 detection assays validated in cancer (TRAP-ddPCR for serum RNA and hybridisation capture for plasma cell-free DNA) can be deployed [11, 12]. The simplest starting point is RT-qPCR (reverse-transcription quantitative PCR) on RNA from patient white blood cells (PBMCs) and serum EVs, stratified by viral trigger. Existing ME/CFS EV sequencing data (Seifert et al. 2026 [14]) could be re-analysed for HSAT2 without new samples. Finding elevated HSAT2 in EVs from patients with documented herpesvirus reactivation, co-varying with MDSC expansion, would support the hypothesis; absence in a well-powered cohort would argue strongly against it.

Limitations. Each step is documented: herpesviruses induce HSAT2 [5], HSAT2 RNA is packaged into EVs in cancer [3], and HSAT2 is detectable in serum [11, 15]. However, the chain linking all three—viral centromere disruption to circulating EV HSAT2 in a non-cancer context—has not been demonstrated. This is a framework for testing, not established fact.

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